SE(3)-WAM: World–Action Modeling at Action Time for Dynamic Manipulation

Anonymous CVPR submission

Figure placeholder: Fig. 1 teaser — forecasting the scene at action time (real robot)

Layout: full width, two parts. **(a) Left, ∼62%:** one real-robot frame of the *conveyor grasp* at the instant a chunk is issued. Overlays in the paper's semantic colors: blue = the target's past SE(3) poses from the causal tracker (small triads at $-350 \ldots 0$ ms, fading with age); orange = target SE(3) triads at the commands' application times, the one at the first application time τ^{a} emphasized, taken from the training-time world head evaluated offline; light orange arrows = coarse 3D flow on the context on the 14×14 grid only; purple = the commanded gripper path meeting the target at τ^{a}. A gray ghost marks the pose observed at capture, with a dashed red arrow showing where a reactive policy would close. A thin timeline strip under the image: capture $\tau_t \rightarrow$ latency $\ell_t \rightarrow$ application τ^{a}. **(b) Right, ∼38%:** top: "Image-space WAM: future frames or latents" (filmstrip icon); middle: "SE(3)-WAM: SE(3) for the target, 3D flow for the context, indexed by application time" (triads + sparse arrows); bottom **(required)**: SR vs. added latency (0–200 ms) for SE(3)-WAM and the strongest baseline (data: the latency sweep of Appendix E.2). **Rules:** real photo kept unretouched; overlays drawn from logged tracker/forecast/command data of that trial; caption states the trial's outcome.

Figure 1. **Forecasting the scene at action time.** Between image capture and the first command, a part on the conveyor moves farther than the grasp tolerates. SE(3)-WAM encodes the target's metric motion (blue) and is trained to forecast the target's SE(3) pose at each command's application time, together with a coarse 3D flow of the surrounding scene (orange, predicted only during training), so that its commands meet the target there (purple). Image-space world–action models forecast appearance instead. Right: success against added latency on Dynamic-14.

Abstract

A robot that grasps a part from a conveyor acts on an image that is a few hundred milliseconds old by the time its command takes effect. Reactive policies ignore this delay, and world–action models that imagine future frames spend their capacity on appearance, at a cost in latency that adds to the delay. We present SE(3)-WAM, a world–action model that forecasts the scene for the instants at which its commands apply. It encodes the target's metric motion from a causal pose tracker, forecasts the target's SE(3) trajectory and a coarse 3D flow of the rest of the scene at physical horizons, and generates commands conditioned on their lead, the time from image capture to their application. The forecast supervises the model during training, and deployment runs only the action stream. For real robots, we pretrain a base on DROID lifted automatically to metric 4D labels. On Dynamic-14, fourteen simulated dynamic tasks evaluated in real time, SE(3)-WAM succeeds in [XX.X]% of held-out episodes against [XX.X]% for the strongest baseline, and on a real conveyor it succeeds in [XX]% of 180 trials against [XX]% for $\pi_{0.5}$. Our analysis finds that the target's metric motion history matters more than visual history, and that supervising the forecast at the commands' application times retains its benefit without generating it at test time.

1. Introduction

Picking a part from a conveyor, placing an object into a moving bin, and catching a ball that rolls off a ramp all require a command that fits the scene at the moment it takes effect. Between image capture and gripper closure, the policy runs inference, commands wait in a queue, and the arm moves. A part on a conveyor at 0.10 m/s travels 15 mm during a 150 ms delay, which is more than the [X] mm that a parallel-jaw grasp of it tolerates. Most visuomotor policies, from diffusion policies to vision–language–action (VLA) and world–action models (WAMs) [10, 29, 39, 46], were developed for quasi-static scenes, where this delay hardly matters [1, 12, 43], and they close behind a moving target (Fig. 1).

Recent dynamic policies address the problem from three sides. Execution schemes shorten or hide the delay by streaming actions and dropping stale ones [6, 34, 43], but some delay always remains. History-based policies give the model stacked frames, optical flow, or kinematic descriptors of the past [12, 22, 28], from which it must infer how the scene moves. World–action models forecast future frames, latents, depth, or semantics and learn actions together with them [33, 39, 46, 52]. Their forecasts are indexed by frames rather than by the times at which commands apply, they model every pixel at the same fidelity, and their size adds to the delay that they are meant to bridge.

We start from what an interaction needs. Whether a grasp or a placement succeeds depends on the target's pose at the instant the command applies, to within a few millimeters, whereas the rest of the scene matters only through the clearance it leaves. The error of a command therefore grows with the target's speed times the lead, and a rotation error grows with the distance from the target's origin to the contact. This suggests three design choices. A policy should *remember* the target's metric motion rather than its appearance, *predict* the target's SE(3) trajectory precisely and the rest of the scene only coarsely, and *act* on forecasts and commands indexed by their application times.

We build SE(3)-WAM on these choices (Fig. 2). A transformer encodes the current RGB-D frame, the robot state, and the target's SE(3) history from a causal pose tracker. A world stream forecasts the target's SE(3) trajectory and a coarse 3D scene flow at physical horizons, and an action stream generates commands that are conditioned on their lead. The world stream shapes the shared representation during training, and deployment runs the action stream alone, which keeps the policy fast. For real robots, we lift DROID [19] to metric 4D labels with off-the-shelf perception [16, 17, 30] and pretrain a base model on it.

We evaluate SE(3)-WAM on Dynamic-14, fourteen simulated tasks with moving targets, receptacles, obstacles, and impacts, on DOMINO [12] and LIBERO [24], and on a real conveyor, with a real-time protocol in which the scene keeps moving while each method computes. SE(3)-WAM reaches [XX.X]% success on held-out Dynamic-14 scenes, [X.X] points above the strongest baseline, improves on DOMINO by [X.X] points, and stays close to strong policies on static LIBERO tasks. On the conveyor, it succeeds in [XX]% of the trials against [XX]% for $\pi_{0.5}$, and its advantage grows with the belt speed. In controlled comparisons, the target's SE(3) history helps once the lead exceeds what an uncompensated grasp tolerates, whereas RGB, flow, and video-latent histories add little, and an SE(3) target forecast with a coarse flow is as useful as image-space forecasts at a fraction of their training cost. When the forecast is supervised at the commands' application times, most of its benefit remains without generating it at test time, as Fast-WAM [48] found for static scenes. In summary, we contribute

- SE(3)-WAM, a world–action model that forecasts the target's SE(3) trajectory and a coarse 3D scene flow at the commands' application times and generates lead-conditioned actions, with a real-robot base pretrained on DROID with lifted 4D labels,
- Dynamic-14 with a real-time protocol, on which SE(3)-WAM improves over reactive, dynamic, and world–action policies, and which we will release with the DROID labels and our code, and
- a controlled analysis of what a dynamic policy needs to remember, to predict, and to read at test time.

2. Related Work

Dynamic manipulation. Compensating delay by prediction is classical in control [32] and predictive tracking [3]. Robotic catching and hand–eye systems estimate an object's trajectory from vision, usually with Kalman-type filters [4], and intercept it [2, 5, 14]. Visual servoing closes the loop on image features [18], and dynamic grasping couples motion prediction with reachability [1]. Among learned policies, DOMINO introduces a dynamic benchmark together with PUMA, which reads historical flow and is supervised with future object features [12]. DynamicVLA streams actions and drops stale ones [43], DynamicWAM conditions a WAM on flow and kinematic descriptors [28], and others predict salient future tokens [33] or event timing [53]. Several recent preprints pursue related ideas with motion-aware world models, velocity fields, or synthesized dynamic demonstrations [9, 21, 49]. For execution, real-time chunking keeps control smooth under delay [6], VLASH rolls the robot state forward to the execution time [34], and the concurrent Chronos Policy conditions on observation and execution timestamps [37]. SE(3)-WAM shares the use of execution time with the last two, but it also forecasts the scene for that time and represents the target's motion in metric terms instead of inferring it from pixels.

World–action models. WAMs couple future prediction with action generation through predictive features [15], visual subgoals [54], or joint video–action generation [20, 39, 46, 57]. DreamVLA forecasts dynamic regions, depth, and semantic features with block-wise attention [52], and others add further supervision to the forecast [47] or compress long histories [26]. Fast-WAM finds that test-time imagination can be skipped without loss on its static tasks [48]. These models forecast in image or latent space and index the future by frames. SE(3)-WAM forecasts a metric, object-level state at physical horizons that match the commands' leads, and we find that in dynamic scenes, too, the benefit of the forecast survives without test-time imagination.

Structured 3D and 4D motion. Rigid-body structure has long been used in learned dynamics models and scene flow [7, 8, 35, 45], and 3D point tracking follows pixels through space [42]. Point tracks and 3D flow have served as policy interfaces [41, 44, 56], point clouds as policy inputs [50], and object poses as structure for world models and policies [11, 27]. PointWorld predicts action-conditioned 3D point flow at scale [16], and feed-forward models reconstruct scene geometry [38] and dynamic point tracks [51]. We use rigid structure only for the target and keep the rest of the scene as a coarse 3D flow, both indexed by the commands' application times.

3. SE(3)-WAM

SE(3)-WAM (Fig. 2) is a transformer with three groups of tokens. Condition tokens encode the current observation and the target's motion history, world queries forecast the scene at physical horizons, and action tokens generate lead-conditioned commands. We predict the world deterministically and generate actions with flow matching [23]. We first state the setting that motivates this design.

3.1. Acting in Physical Time

Application times. We consider a policy that receives images and robot states together with their capture times. At query t, its history is $\mathcal{H}_t = \{(o_i, s_i, \tau_i)\}_{\tau_i \le \tau_t}$, and it outputs a chunk $A_t = (\mathbf{a}_{t,0}, \dots, \mathbf{a}_{t,H-1})$ whose j-th command takes effect at

$$\tau^{\mathrm{a}}_{t,j} = \tau_t + \ell_t + j\Delta_a, \tag{1}$$

where ℓ_t is the latency from capture to the first application (preprocessing, inference, queuing, and actuation) and Δ_a is the control period. We call $\Delta_{t,j} = \tau^{\mathrm{a}}_{t,j} - \tau_t$ the *lead* of command j, which must fit the scene at $\tau^{\mathrm{a}}_{t,j}$ rather than at τ_t (Fig. 3).

Target and context. Let $W_t(\mathbf{x}, \tau) \in \mathbb{R}^3$ be the displacement between τ_t and τ of the scene point at $\mathbf{x}$, in the robot base frame. The *target* $\mathcal{O}$ is the rigid body whose pose error enters a contact or containment tolerance ε, e.g., an object to grasp or a receptacle to place into. The *context* $\mathcal{C}$ contains the bodies that only need to be avoided—a belt or an occluder—with clearance m. A body can play both roles, e.g., a wire whose entry must be met precisely and whose loop must be avoided. On the target, the motion field is set by its pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$:

$$W_t(\mathbf{x}, \tau) = T(\tau)\, T(\tau_t)^{-1}\mathbf{x} - \mathbf{x}, \qquad \mathbf{x} \in \mathcal{O}. \tag{2}$$

Where the error comes from. Let a policy act, explicitly or implicitly, on an estimate $\hat{T}$ of the target pose at τ^{a}. For a contact point at lever arm r and a rotation error θ, the contact error is at most $\|\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}})\| + 2r\sin(\theta/2)$. Writing the motion over the lead Δ as constant velocity $\mathbf{v}$ plus a residual $\boldsymbol{\rho}$ (acceleration, rolling, impacts) gives

$$\|\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}})\| \le e_{\mathrm{loc}} + \hat{\Delta}\, e_v + \|\mathbf{v}\|\, e_\tau + e_{\mathrm{dyn}}, \tag{3}$$

the errors of the current position, of the velocity times the lead, of the timing times the speed, and of the residual motion (Appendix G). For the context, a collision requires a prediction error of at least m somewhere along the commanded path. A policy that ignores motion therefore misses once $\|\mathbf{v}\|\Delta$ exceeds ε, while the context can be predicted far more coarsely. These observations motivate the three parts of SE(3)-WAM, which rely on a pose tracker for the current position (e_{loc}): a metric motion history, which addresses the velocity error e_v (Sec. 3.2); an SE(3) forecast of the target with a coarse context flow, which addresses e_{dyn} and rotation errors (Sec. 3.3); and commands that know their lead and are trained with forecasts at their application times, which address e_τ (Sec. 3.4).

3.2. Metric Target-Motion History

We estimate the target pose in every RGB-D frame at full resolution with a causal tracker. SAM 2 propagates a mask from a prompt on the target [30], and FoundationPose tracks the object from a CAD model or a few reference views [40]. Without either, we use a monocular 3D detector that estimates a metric box pose in every frame [17]. Each estimate is emitted once, with its capture time and a validity flag, and later frames do not revise it. We resample this causal track at $-350, -300, \dots, 0$ ms, so that its meaning does not depend on the camera rate, and build 8 motion tokens with interval motions, finite-difference velocities and accelerations, and the current pose, in the base frame and in physical units (Appendix A.1). A frozen MAE ViT-B/16 [13] encodes the current RGB image into 196 patch tokens, and the depth image gives each patch a metric 3D position for rotary attention. Robot tokens hold 8 end-effector states and the pose commanded for the first application time. The model receives no past images, and we train it on the same causal pose estimates that it sees at test time, tracking failures included.

3.3. Role-Dependent 4D Forecast

The world queries forecast W_t at horizons $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50$ ms and $h = 1, \dots, 8$, which cover the leads of all commands executed before the next replan, so that the forecast is supervised at the times when the commands apply (Appendix G.4). We parameterize the target through Eq. (2). For each horizon,

$$\mathbf{d}_h = \mathbf{p}(\tau_t{+}\delta_h) - \mathbf{p}(\tau_t), \quad \boldsymbol{\phi}_h = \mathrm{Log}\big(R(\tau_t{+}\delta_h)R(\tau_t)^\top\big)^\vee. \tag{4}$$

An object query regresses $(\mathbf{d}_h, \boldsymbol{\phi}_h)$ with a squared translation error plus λ_R times the squared geodesic rotation error ($\mathcal{L}_P$). For rotationally symmetric targets, we exclude the unobservable rotation about the symmetry axis. We represent the rest of the scene as a 3D scene flow, the base-frame displacements $F_h \in \mathbb{R}^{N\times 3}$ of $N = 196$ points on a fixed 14×14 image grid, about [X] mm apart in the workspace. One scene query per point, with an embedding of its role (target, robot, or context), regresses the point's displacements and visibility ($\mathcal{L}_F$). The grid is deliberately coarse, since the context is tolerated up to its clearance. Future depth, DINOv3 features [31], and the RGB of moving surfaces serve as additional training targets for the scene queries ($\mathcal{L}_{\mathrm{aux}}$). All labels are cumulative from the current state, and losses use valid labels only.

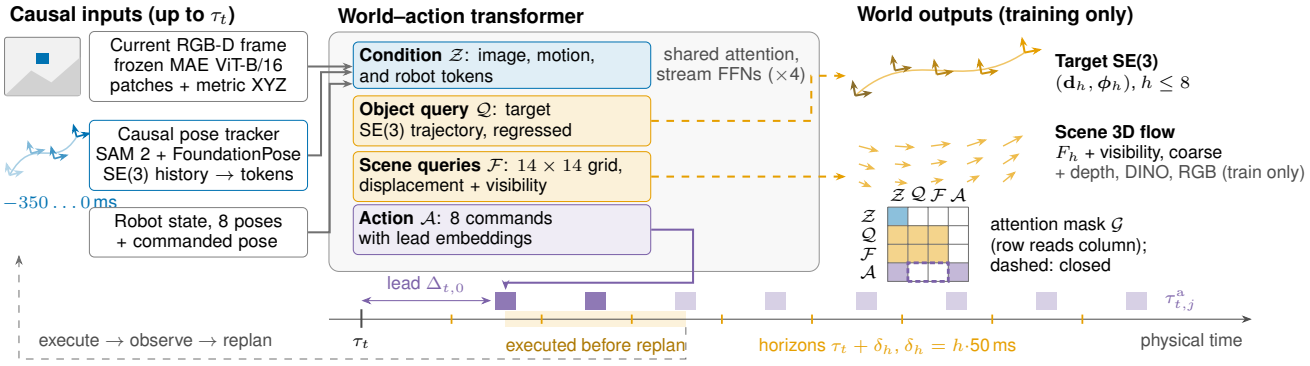


Figure 2. **SE(3)-WAM.** Condition tokens $\mathcal{Z}$ encode the current RGB-D frame, the target's SE(3) history from a causal tracker, and the robot state. An object query $\mathcal{Q}$ and scene queries $\mathcal{F}$ regress the target's SE(3) trajectory and a coarse 3D scene flow with visibility at physical horizons $\tau_t + \delta_h$, and appearance heads add training targets. Action tokens $\mathcal{A}$ generate lead-conditioned commands that apply at $\tau^{\mathrm{a}}_{t,j}$, with replanning every 100 ms. The mask $\mathcal{G}$ sets which groups each group attends to. Action tokens attend to no world token, so deployment runs the action stream alone.

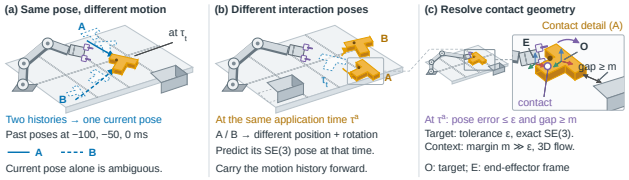


Figure 3. **Acting in physical time** (schematic). (a) Two histories reach the same pose at capture time τ_t but imply different motions. (b) At the application time τ^{a}, they lead to different target poses. (c) A contact requires the end-effector frame E to match the target O within ε, whereas the context matters only through the gap it leaves, $m \gg \varepsilon$.

3.4. Lead-Conditioned Actions

A chunk holds 8 commands that are applied at the times of Eq. (1), with a control period Δ_a of 50 ms (20 Hz). With replanning every 100 ms, the first two commands of a chunk are executed before the next chunk replaces the rest. Each command is a base-frame translation and a left-relative rotation of the end effector, plus a gripper command, taken relative to the pose commanded for the preceding application time, which the controller knows [34]. Each action token embeds its lead $\Delta_{t,j} = \ell_t + j\Delta_a$. During training, we sample $\ell \sim \mathcal{U}[0, 250\,ms]$ and read the command labels from the demonstration at $\tau_t + \ell + j\Delta_a$, so that a single model serves any latency in this range. For a normalized chunk $\mathbf{a}_1$, noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and $u \sim \mathcal{U}[0, 1]$, the action loss is

$$\mathcal{L}_A = \mathbb{E}\left\| v_\psi(\mathbf{a}_u, u \mid \mathcal{Z}_t, \hat{\mathcal{W}}_t, \mathcal{G}) - (\mathbf{a}_1 - \mathbf{a}_0) \right\|^2, \quad (5)$$

with $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, condition tokens $\mathcal{Z}_t$, and the world tokens $\hat{\mathcal{W}}_t$ that the coupling lets the action read (none in R0 and R1). The total loss is $\mathcal{L} = \mathcal{L}_A + \lambda_F\mathcal{L}_F + \lambda_P\mathcal{L}_P + \lambda_{\mathrm{aux}}\mathcal{L}_{\mathrm{aux}}$.

World–action coupling. The attention mask $\mathcal{G}$ acts on four token groups: the condition $\mathcal{Z}$, the object query $\mathcal{Q}$, the scene queries $\mathcal{F}$, and the actions $\mathcal{A}$. Condition tokens attend only to each other, world queries attend to the condition and to each other, and action tokens attend to the condition, to themselves, and to the part of the world that the coupling allows. The action reads no world token. The world losses shape the shared parameters during training, and deployment runs only the action stream. The two streams share the attention projections and have separate feed-forward networks. Sec. 4.4 compares this choice with the alternatives, which we label as follows: R0 disables the world losses, R1 lets the action read no world token, R2 lets it read the hidden states of the world queries, R3 lets the world read the noisy actions, R4 opens both directions, and R5 encodes the decoded geometry for the action (Appendix A.5).

Execution and implementation. At test time, we set the leads from the executor's latency estimate, the recent p95 inference time plus the actuation delay. The transformer has 4 layers of width 768 with 8 heads, 47.3M backbone parameters, and 62.2M trainable parameters in total. The image encoder and the DINOv3 teacher stay frozen but run at every training step, and they dominate its cost. On Dynamic-14, we train for 4 epochs (about 40k updates at a global batch size of 16) with data parallelism, which takes about 7.7 hours per training seed on eight RTX PRO 6000 GPUs. At test time, the action stream integrates 5 Euler steps in half precision, with a p95 time-to-action of [XX] ms on an RTX 5090 (Appendices A.6 and D.4).

3.5. Pretraining on Real Robot Data Lifted to 4D

Real robot data provide neither object poses nor scene flow, so we lift DROID [19] to metric 4D labels without object

models. The scene flow comes from the 3D tracks and optimized camera calibrations that PointWorld released for DROID [16]. We take the object named in the instruction as the target. An open-vocabulary detector finds it [25], SAM 2 tracks its mask, and the monocular 3D detector estimates its metric pose. Inputs come from a forward track, as at deployment, and the labels of Eq. (4) from a smoothed forward–backward track over the whole episode. DROID's targets move mostly when the robot moves them ([X]% of windows show target motion before contact), so pretraining mainly teaches metric geometry, 3D flow, and their coupling with lead-conditioned actions (Fig. S1). We initialize the real-robot policy from this base. The simulation model is trained from the generic initialization, a frozen MAE encoder with a randomly initialized transformer, and a controlled comparison post-trains the base on Dynamic-14 (Appendix B).

4. Experiments

We evaluate SE(3)-WAM against reactive, dynamic, and world–action policies in simulation and on a real robot (Secs. 4.2 and 4.3) and then analyze which of its components matter and why (Sec. 4.4).

4.1. Setup

Dynamic-14. We built fourteen tasks in MuJoCo/robosuite [36, 58] in five families: grasping a moving target (harmonic, conveyor, translating, rotating, and occluded motion), placing into a moving receptacle, passing a wand along a rotating wire, capturing rolling or sliding objects, and grasping after an impact (Appendix C, Fig. S3). Sweeps vary the speed (0.5–1.5×), the added latency (0–200 ms), the passage clearance ([values] mm), and the pose noise. Scripted experts with simulator state provide 800 demonstrations per task. We select configurations on development scenes and report every Dynamic-14 number on 1,400 held-out scenes (100 per task, new seeds). Our variants average three training seeds, and each baseline is trained once. We give paired 95% intervals from a bootstrap over held-out scenes. In simulation, the tracker is prompted with the rendered mask of the target in the first frame and uses the simulator mesh as its object model.

Protocol. All methods run in real time under the same executor, and physics advances while a method computes, by its measured latency on one RTX 5090. A new chunk is requested every 100 ms, or when the previous inference ends, commands execute at their application times, and stale commands are dropped. The +150 ms setting adds 150 ms of delay for every method, and the paused setting stops physics during inference (Appendix E.2). We report the task-balanced success rate (SR), the contact-time error e_t, the position error at first contact e_p, and the p95 time-to-action. On DOMINO [12], we evaluate single-arm interception and tracking, and LIBERO [24] checks that the structured world does not cost performance on static tasks.

Baselines. We compare with reactive policies (ACT [55], Diffusion Policy [10], and $\pi_{0.5}$ [29]), dynamic VLAs (DynamicVLA [43] and PUMA [12]), image- or latent-space WAMs (DreamVLA [52] and OpenWAM-α [39]), and a dynamic WAM (DynamicWAM [28]). Each baseline uses its official code, inputs, and recipe, is trained on the same demonstrations, and receives a rendered wrist camera if its recipe uses one (Appendix D). SE(3)-WAM uses only the agent view, but it also uses depth and the tracked target pose. To separate these inputs from the model, $\pi_{0.5}$ and DynamicWAM also receive our SE(3) history tokens (+SE(3)), and we report SE(3)-WAM with poses from the monocular 3D detector, which needs no object model. Two baselines use the tracker alone. *CV input* is SE(3)-WAM without world supervision that receives the constant-velocity extrapolation of the tracked pose as input tokens, and a Kalman interceptor plans to the target pose predicted by a constant-acceleration Kalman filter [4].

4.2. Main Results

In real time, SE(3)-WAM reaches [XX.X]% SR on held-out Dynamic-14 scenes (Table 1), [X.X] points above the strongest baseline, [name of the strongest baseline], and [X.X] points above the strongest baseline that receives the same SE(3) history, [name of the strongest +SE(3) baseline]. The margin is largest on [rotating targets and on fast conveyor and translating targets], where the target moves farthest during the lead[, and no method solves the SE(3) placement into the rotating receptacle]. With 150 ms of added delay, SE(3)-WAM keeps [XX.X]%, whereas reactive policies drop to [XX.X]% and image-space WAMs to [XX.X]%. With physics paused during inference, the difference to [name of the strongest paused baseline] is [±X.X] points, which separates the contribution of speed from that of the model. Detector poses instead of an object model cost [X.X] points, and the CV-input variant and the Kalman interceptor trail SE(3)-WAM by [X.X] and [X.X] points, mostly on non-constant motion (Appendix F.3). On DOMINO, SE(3)-WAM obtains [XX.X]% against [XX.X]% for [name of the strongest DOMINO baseline], and on static LIBERO tasks it is within [X.X] points of $\pi_{0.5}$ ([XX.X]% vs. [XX.X]%).

4.3. Real-Robot Experiments

A seven-DoF Franka arm with a parallel-jaw gripper must grasp an object that a belt conveyor carries past it, lift it, and hold it for 0.5 s, observed by a calibrated RGB-D camera (Fig. S4). The three objects differ in how much their orientation matters: a box with unequal sides, a slender can, and a cup with one handle. The belt runs at 0.04, 0.08, or 0.12 m/s.

Table 1. **Main results.** Held-out SR (%) on Dynamic-14 (1,400 new scenes, task-balanced, real time, three training seeds for our variants and one for baselines) by family, with 150 ms of added delay, and with physics paused during inference, on single-arm DOMINO and LIBERO (four suites), and p95 time-to-action on one RTX 5090. Inputs: R, RGB and robot state. D, depth. P, target SE(3) history from our tracker with an object model. P′, from a monocular 3D detector. Best in bold. Intervals in Appendix E.

		Dynamic-14, real time unless noted										
Method	Inputs	All	Grasp	Place	Pass	Capture	Impact	+150 ms	Paused	DOMINO	LIBERO	p95 (ms)
ACT [55]	R	—	—	—	—	—	—	—	—	—	—	—
Diffusion Policy [10]	R	—	—	—	—	—	—	—	—	—	—	—
$\pi_{0.5}$ [29]	R	—	—	—	—	—	—	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	R+P	—	—	—	—	—	—	—	—	—	—	—
DynamicVLA [43]	R	—	—	—	—	—	—	—	—	—	—	—
PUMA [12]	R	—	—	—	—	—	—	—	—	—	—	—
DreamVLA [52]	R	—	—	—	—	—	—	—	—	—	—	—
OpenWAM-α [39]	R	—	—	—	—	—	—	—	—	—	—	—
DynamicWAM [28]	R	—	—	—	—	—	—	—	—	—	—	—
DynamicWAM + SE(3)	R+P	—	—	—	—	—	—	—	—	—	—	—
SE(3)-WAM, CV input	RD+P	—	—	—	—	—	—	—	—	—	—	—
Kalman interceptor	D+P	—	—	—	—	—	—	—	—	–	–	—
SE(3)-WAM, detector poses	RD+P′	—	—	—	—	—	—	—	—	—	—	—
SE(3)-WAM	RD+P	—	—	—	—	—	—	—	—	—	—	—

Table 2. **Real-robot conveyor grasp.** Success rate (%) by object and by belt speed (slow, mid, fast: 0.04, 0.08, 0.12 m/s), 60 trials each, and over all 180 trials of a method. Every method is fine-tuned with three seeds, one per block of 60 trials. Binomial 95% intervals are given in Appendix D.3.

	Object			Belt speed			
Method	Box	Can	Cup	Slow	Mid	Fast	All
$\pi_{0.5}$ [29]	—	—	—	—	—	—	—
SE(3)-WAM, no DROID base	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—

Figure placeholder: Fig. 4 — real conveyor trials

Rows: a success of SE(3)-WAM on the cup at 0.12 m/s; $\pi_{0.5}$ on the same trial key; a failure of SE(3)-WAM. **Columns:** the query frame, the first application time τ^{a}, and contact. **Overlays:** blue tracked past poses; orange forecast triads at the commands' application times; purple commanded gripper path. Real photos, unretouched; overlays from logged data of each trial.

Figure 4. **Real conveyor trials.** SE(3)-WAM closes where the cup will be when the command applies, whereas $\pi_{0.5}$ closes behind it. Bottom: a failure after the tracker lost the handle under the gripper.

Each method is fine-tuned on the same [300] teleoperated demonstrations with three seeds and runs 180 trials, 20 per object and speed, in three blocks of shared trial keys with one seed per block. SE(3)-WAM uses [FoundationPose with a CAD model or reference views of each object], whereas its base was pretrained on detector poses. SE(3)-WAM succeeds in [XX.X]% of its trials (95% interval [[a, b]]), against [XX.X]% for $\pi_{0.5}$ and [XX.X]% without the DROID base (Table 2). Its advantage over $\pi_{0.5}$ grows from [X.X] points at 0.04 m/s to [X.X] at 0.12 m/s and is largest on [the cup] (Table S10, Fig. 4). Its remaining failures are mostly [late closures after tracking losses under the gripper].

Pretraining. The DROID base raises the real-robot SR by [X.X] points over the generic initialization (Table 2). In simulation, post-training the base on Dynamic-14 changes SR by [X.X] points with all demonstrations and by [X.X] with 10% of them. At pilot scale, a base pretrained with actions only recovers [XX]% of the low-data gain of a world–action base with the same budget, so most of the benefit comes from the 4D labels (Appendix B).

4.4. Analysis: What Does a Dynamic Policy Need?

We now ask which parts of SE(3)-WAM produce these gains. Each variant changes one component of the final model and keeps the data, the training budget, and the other components fixed (Table 3 and Fig. 5). The complete comparisons are in Appendix F.

What to remember. We hold the current frame, the current target pose, and the robot history fixed. Adding eight frames of RGB, optical flow over the same window with DynamicWAM's motion statistics [28], or video-VAE latents changes SR by [X.X] to [X.X] points. The causal SE(3)

Figure placeholder: Fig. 5 — analysis of the components (four panels)

(a) History. x: added latency (0–200 ms); y: ΔSR vs. current frame. Lines: RGB window, optical flow, video-VAE latent (gray), SE(3) causal (blue), detector SE(3) (blue dotted), SE(3) oracle (blue dashed); stationary targets as a faint line; ticks at the families' critical leads $\varepsilon/\|\mathbf{v}\|$. **(b) Target forecast.** Bars by motion type (constant velocity, harmonic, rolling/sliding, post-impact): forecast error at τ^{a} of CV, CA, and the learned SE(3) forecast (orange), with SR of the corresponding policies as dots. **(c) Context.** Passage tasks only. x: clearance (mm); y: SR; lines: no context flow, 14^2, 28^2, 56^2 grids, future latent; markers: each grid's measured path error. **(d) Coupling.** x: added latency; y: ΔSR vs. the final model; lines: the other couplings and world losses off (R0, dotted); stationary targets as points at the right. 95% paired bootstrap bands, held-out scenes, three seeds.

Figure 5. **What the components contribute.** (a) The SE(3) history helps once the lead exceeds the critical lead, and more with latency, while visual histories stay flat. (b) The learned SE(3) forecast beats extrapolation on non-constant motion. (c) A coarse context flow suffices until the clearance reaches its path error. (d) Training-only world supervision matches online use of the forecast at every latency. Bands show 95% paired bootstrap intervals.

Table 3. **Component analysis** on held-out Dynamic-14 scenes (three training seeds). Each block changes one component of SE(3)-WAM (⋆). SR, SR_{150}: success (%) without and with 150 ms of added delay. e_t: contact-time error (ms). e_p: position error at first contact (mm). p95: time-to-action (ms).

Variant	SR	SR_{150}	e_t	e_p	p95
SE(3)-WAM ⋆	—	—	—	—	—
History (current frame and pose kept)					
none	—	—	—	—	—
RGB window (8 frames)	—	—	—	—	—
optical flow, motion statistics	—	—	—	—	—
video-VAE latent	—	—	—	—	—
SE(3), oracle	—	—	—	—	—
Forecast					
none (world losses off, R0)	—	—	—	—	—
target tracks + flow	—	—	—	—	—
target SE(3) only	—	—	—	—	—
SE(3) + 28 × 28 flow	—	—	—	—	—
SE(3) + future RGB/latent (no flow)	—	—	—	—	—
no appearance targets	—	—	—	—	—
Coupling					
action reads world states (R2)	—	—	—	—	—
bidirectional world states (R4)	—	—	—	—	—
action reads decoded geometry (R5)	—	—	—	—	—
Physical time					
no lead embedding	—	—	—	—	—
leads in control steps	—	—	—	—	—
horizons off by 50 ms	—	—	—	—	—

history adds [X.X] points and the oracle history [X.X] (Table 3). The gain appears once the lead exceeds the critical lead $\varepsilon/\|\mathbf{v}\|$ of a family, grows with latency, and vanishes for stationary targets (Fig. 5a). Frozen readouts show that visual histories do contain the motion (Appendix F.2), but at our data scale the policy does not turn it into metric velocity, even with metric XYZ for past frames ([±X.X] points, Table S17). Pose noise and occlusions are analyzed in Appendices F.2 and G.5. The same history given to $\pi_{0.5}$ and DynamicWAM raises their SR by [X.X] and [X.X] points (Table 1), so the benefit is not specific to our model.

What to predict. Supervising the target's SE(3) trajectory instead of point tracks changes SR by [±X.X] points overall but adds [X.X] on rotating targets, where the lever arm amplifies rotation errors. At the application times, the learned SE(3) forecast is as accurate as constant-velocity and constant-acceleration extrapolation on conveyor motion, but [X] times more accurate on rolling, sliding, and post-impact motion (Fig. 5b), although before an impact it averages the possible outcomes (Fig. 6). For the context, the 14×14 flow, with a path error of [X] mm, matches 28^2 and 56^2 grids until the passage clearance falls to [X] mm (Fig. 5c, Table S21), at [X] times lower training cost for the world stream. Removing the context flow costs [X.X] points, mostly on passage tasks. As the predicted world, future RGB, VAE latents, depth, DINO features, or latents of a frozen video model add at most [X.X] points to the SE(3) trajectory at matched capacity, at up to [X] times the training cost. Appearance helps in another role: as auxiliary targets of the scene queries, whose heads are dropped at deployment, depth, DINO features, and moving-region RGB add [X.X] points.

How to use the forecast. Training-only world supervision with action-only deployment performs on par with the couplings that read the world ([±X.X] points at 0 ms and [±X.X] at 150 ms, Fig. 5d) at [X.X] times lower time-to-action. Disabling the world losses costs [X.X] points, and this gap grows with the lead and vanishes for stationary targets. The forecast thus helps as a training signal for lead-conditioned actions, and generating it at test time adds latency without adding success.

Physical time. Removing the lead embedding costs [X.X] points, and [X.X] under 150 ms of added delay. Counting

Figure placeholder: Fig. 6 — forecasts at application time

Rows: a rolling capture (median forecast error) and a post-impact grasp that fails. **Columns:** the query frame at τ_t, the frame at τ^a, and a 3D inset in the base frame. **Overlays:** blue past poses, orange forecast triads and coarse flow arrows, gray ground truth at the same times, purple commanded path. Full set in Fig. S5.

Figure 6. **Forecasts at the application time** on held-out Dynamic-14 episodes. Top: the SE(3) forecast follows a rolling target that extrapolation overshoots. Bottom: before an impact, the regressed forecast averages the possible outcomes, and the grasp misses.

leads in control steps from the first command costs [X.X] points when the test latency differs from the training latency, and supervising the forecast away from the application times by 50 ms costs [X.X] points on the fastest third of the scenes but [X.X] for stationary targets (Appendix F.5).

5. Conclusion

We presented SE(3)-WAM, a world–action model for manipulating moving objects that forecasts the scene for the instants at which its commands apply. It remembers the target's metric motion, forecasts the target's SE(3) trajectory precisely and the rest of the scene coarsely, and conditions each command on its lead. In real time, it improves on reactive, dynamic, and world–action policies in simulation, on DOMINO, and on a real conveyor, and a base pretrained on DROID with lifted 4D labels carries over to the real robot. Our analysis attributes the gains to the metric motion history and to lead-conditioned actions learned together with forecasts at their application times, rather than to richer appearance models. For world–action models, it favors supervising the future at the instants when commands apply and keeping the deployed policy fast, over generating the future at test time.

Limitations. SE(3)-WAM assumes a rigid target that moves on its own until contact, so deformable or articulated targets and post-contact dynamics are outside its scope. It depends on a pose tracker whose failures propagate to the policy. Poses from a monocular 3D detector remove the need for an object model but cost accuracy, and the regressed target forecast cannot represent multimodal futures, e.g., before an impact. Lifting DROID to 4D labels costs substantial computation. Finally, our tasks have a single target and slow context motion, and the real-robot study covers one task on one platform.

LLM Usage Statement

[To be completed before submission: actual tools used, affected parts of the research and manuscript, and extent of use.]

References

[1] Iretiayo Akinola, Jingxi Xu, Shuran Song, and Peter K. Allen. Dynamic grasping with reachability and motion awareness. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 9422–9429, 2021. 1, 2

[2] Peter K. Allen, Aleksandar Timcenko, Billibon Yoshimi, and Paul Michelman. Automated tracking and grasping of a moving object with a robotic hand-eye system. *IEEE Transactions on Robotics and Automation*, 9(2):152–165, 1993. 2

[3] Ronald Azuma and Gary Bishop. Improving static and dynamic registration in an optical see-through HMD. In *Proceedings of SIGGRAPH*, pages 197–204, 1994. 2

[4] Yaakov Bar-Shalom, X. Rong Li, and Thiagalingam Kirubarajan. *Estimation with Applications to Tracking and Navigation*. Wiley, 2001. 2, 5

[5] Berthold Bäuml, Thomas Wimböck, and Gerd Hirzinger. Kinematically optimal catching a flying ball with a hand-arm-system. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2010. 2

[6] Kevin Black, Manuel Y. Galliker, and Sergey Levine. Real-time execution of action chunking flow policies. In *Advances in Neural Information Processing Systems*, 2025. 1, 2

[7] Arunkumar Byravan and Dieter Fox. SE3-Nets: Learning rigid body motion using deep neural networks. In *IEEE International Conference on Robotics and Automation*, 2017. 2

[8] Arunkumar Byravan, Felix Leeb, Franziska Meier, and Dieter Fox. SE3-Pose-Nets: Structured deep dynamics models for visuomotor control. In *IEEE International Conference on Robotics and Automation*, 2018. 2

[9] Chongkei Chang and Zhidong Deng. DynaWM: A base-VLA-guided world foundation model for moving-object manipulation. *arXiv preprint arXiv:2607.02604*, 2026. 2

[10] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin C. M. Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. 1, 5, 6

[11] David Emukpere, Romain Deffayet, Bingbing Wu, Romain Brégier, Michael Niemaz, Jean-Luc Meunier, Denys Proux, Jean-Michel Renders, and Seungsu Kim. Disentangled object-centric image representation for robotic manipulation. *arXiv preprint arXiv:2503.11565*, 2025. 2

[12] Heng Fang, Shangru Li, Shuhan Wang, Xuanyang Xi, Dingkang Liang, and Xiang Bai. Towards generalizable robotic manipulation in dynamic environments. *arXiv preprint arXiv:2603.15620*, 2026. 1, 2, 5, 6

[13] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16000–16009, 2022. 3

[14] B. Hove and Jean-Jacques E. Slotine. Experiments in robotic catching. In *American Control Conference*, 1991. 2

[15] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. In *International Conference on Machine Learning*, 2025. 2

[16] Wenlong Huang, Yu-Wei Chao, Arsalan Mousavian, Ming-Yu Liu, Dieter Fox, Kaichun Mo, and Li Fei-Fei. PointWorld: Scaling 3D world models for in-the-wild robotic manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 2, 5

[17] Weikai Huang, Jieyu Zhang, Sijun Li, Taoyang Jia, Jiafei Duan, et al. WildDet3D: Scaling promptable 3D detection in the wild. *arXiv preprint arXiv:2604.08626*, 2026. 2, 3

[18] Seth Hutchinson, Gregory D. Hager, and Peter I. Corke. A tutorial on visual servo control. *IEEE Transactions on Robotics and Automation*, 12(5):651–670, 1996. 2

[19] Alexander Khazatsky, Karl Pertsch, Suraj Nair, et al. DROID: A large-scale in-the-wild robot manipulation dataset. In *Proceedings of Robotics: Science and Systems*, 2024. 2, 4

[20] Shuang Li, Yihuai Gao, Dorsa Sadigh, and Shuran Song. Unified video action model. In *Proceedings of Robotics: Science and Systems*, 2025. 2

[21] Haoran Liao, Pengyue Wang, Shuoyu Chen, Kehan Cheng, Xuhang Chen, Yuhao Lin, Mu Lin, Zhizhao Liang, Xiaoyi Fan, Chengyi Xing, Dan Niu, Yi-Lin Wei, and Wei-Shi Zheng. DynamicManip: Enabling dynamic manipulation from a single static demonstration. *arXiv preprint arXiv:2608.01452*, 2026. 2

[22] Minghui Lin, Pengxiang Ding, Shu Wang, Zifeng Zhuang, Yang Liu, Xinyang Tong, Wenxuan Song, Shangke Lyu, Siteng Huang, and Donglin Wang. HiF-VLA: Hindsight, insight and foresight through motion representation for vision-language-action models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 1

[23] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. 3

[24] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2023. 2, 5

[25] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *European Conference on Computer Vision*, 2024. 5

[26] Yushan Liu, Jingjing Fan, Shoujie Li, Yifan Xie, Xiao-Ping Zhang, and Wenbo Ding. PACT-WAM: Predicting actions and visual foresight with compact temporal encoding for robot manipulation. *arXiv preprint arXiv:2602.15882*, 2026. 2

[27] Yushan Liu, Peibo Sun, Shoujie Li, Yifan Xie, Lingfeng Zhang, Xintao Chao, Shiyuan Dong, Fang Chen, Xiao-Ping Zhang, and Wenbo Ding. OA-WAM: Object-addressable world action model for robust robot manipulation. *arXiv preprint arXiv:2605.06481*, 2026. 2

[28] Yunfan Lou, Hewen Gao, Xiyu Zhu, Zhuoran Qiao, Xuan Han, Yifan Yang, Yifan Ye, Boxian Yao, and Zhibo Pang. DynamicWAM: Dual-path motion conditioning for world-action models in dynamic manipulation. *arXiv preprint arXiv:2608.00793*, 2026. 1, 2, 5, 6

[29] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025. 1, 5, 6

[30] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. In *International Conference on Learning Representations*, 2025. 2, 3

[31] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. 3

[32] Otto J. M. Smith. Closer control of loops with dead time. *Chemical Engineering Progress*, 53(5):217–219, 1957. 2

[33] Shahram Najam Syed, Arthur Jakobsson, Haoran Hao, and Jeffrey Ichnowski. Intercepting the future: Latent-space predictive world model for dynamic VLA manipulation. *arXiv preprint arXiv:2606.02486*, 2026. 1, 2

[34] Jiaming Tang, Yufei Sun, Yilong Zhao, Shang Yang, Yujun Lin, Zhuoyang Zhang, James Hou, Yao Lu, Zhijian Liu, and Song Han. VLASH: Real-time VLAs via future-state-aware asynchronous inference. *arXiv preprint arXiv:2512.01031*, 2025. 1, 2, 4

[35] Zachary Teed and Jia Deng. RAFT-3D: Scene flow using rigid-motion embeddings. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021. 2

[36] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5026–5033, 2012. 5

[37] Changhao Wang, Jaden Clark, and Shuran Song. Chronos policy: Learning robot policies on physical time. *arXiv preprint arXiv:2610.13929*, 2026. 2

[38] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual geometry grounded transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2

[39] Yuran Wang, Siqiao Huang, Mingleyang Li, Chenhao Zhang, Jiaqi Liang, Weiyang Jin, Yue Chen, Xuemin Chi, Donghao Zhou, Qize Yu, Yu-Kai Wang, Yuhan Rui, Shenzhe Yao, Zhen Yuan, Zhenhao Shen, Kefei Zhu, Zijie Zhu, Ning Gao, Xiaowei Chi, Guanqi He, Shanghang Zhang, Hao Dong, Lin Shao, and Hang Zhao. OpenWAM: An open, modular exploration towards systematic world-action model pretraining. *arXiv preprint arXiv:2609.07398*, 2026. 1, 2, 5, 6

[40] Bowen Wen, Wei Yang, Jan Kautz, and Stan Birchfield. FoundationPose: Unified 6D pose estimation and tracking of novel objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17868–17879, 2024. 3

[41] Chuan Wen, Xingyu Lin, John So, Kai Chen, Qi Dou, Yang Gao, and Pieter Abbeel. Any-point trajectory modeling for policy learning. In *Proceedings of Robotics: Science and Systems*, 2024. 2

[42] Yuxi Xiao, Qianqian Wang, Shangzhan Zhang, Nan Xue, Sida Peng, Yujun Shen, and Xiaowei Zhou. SpatialTracker: Tracking any 2D pixels in 3D space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. 2

[43] Haozhe Xie, Beichen Wen, Jiarui Zheng, Zhaoxi Chen, Fangzhou Hong, Haiwen Diao, and Ziwei Liu. DynamicVLA: A vision-language-action model for dynamic object manipulation. *arXiv preprint arXiv:2601.22153*, 2026. 1, 2, 5, 6

[44] Mengda Xu, Zhenjia Xu, Yinghao Xu, Cheng Chi, Gordon Wetzstein, Manuela Veloso, and Shuran Song. Flow as the cross-domain manipulation interface. In *Conference on Robot Learning*, 2024. 2

[45] Zhenjia Xu, Zhanpeng He, Jiajun Wu, and Shuran Song. Learning 3D dynamic scene representations for robot manipulation. In *Conference on Robot Learning*, 2020. 2

[46] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, et al. World action models are zero-shot policies. *arXiv preprint arXiv:2602.15922*, 2026. 1, 2

[47] Shanglin Yuan, Weiheng Zhao, Xin Shi, Haoyi Jiang, Xianda Guo, Liu Liu, Wenyu Liu, Wei Sui, and Xinggang Wang. DreamWAM: Beyond RGB future prediction for world action models. *arXiv preprint arXiv:2608.04996*, 2026. 2

[48] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-WAM: Do world action models need test-time future imagination? *arXiv preprint arXiv:2603.16666*, 2026. 2

[49] Peng Yun, Shouwang Huang, Hao Li, Jinxi Li, Jianan Wang, and Bo Yang. PhysMani: Physics-principled 3D world model for dynamic object manipulation. *arXiv preprint arXiv:2607.01938*, 2026. 2

[50] Yanjie Ze, Gu Zhang, Kangning Zhang, Chenyuan Hu, Muhan Wang, and Huazhe Xu. 3D diffusion policy: Generalizable visuomotor policy learning via simple 3D representations. In *Proceedings of Robotics: Science and Systems*, 2024. 2

[51] Chuhan Zhang, Guillaume Le Moing, Skanda Koppula, Ignacio Rocco, Liliane Momeni, Junyu Xie, Shuyang Sun, Rahul Sukthankar, Joëlle K. Barral, Raia Hadsell, Zoubin Ghahramani, Andrew Zisserman, Junlin Zhang, and Mehdi S. M. Sajjadi. Efficiently reconstructing dynamic scenes one D4RT at a time. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 2

[52] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, XinQiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He

Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. DreamVLA: A vision-language-action model dreamed with comprehensive world knowledge. In *Advances in Neural Information Processing Systems*, pages 24195–24228, 2025. 1, 2, 5, 6

[53] Yifan Zhang, Ruiping Wang, and Xilin Chen. Dynamic behavior cloning with temporal feature prediction: Enhancing robotic arm manipulation in moving object tasks. *IEEE Robotics and Automation Letters*, 10(6):5209–5216, 2025. 2

[54] Qingqing Zhao, Yao Lu, Moo Jin Kim, Zipeng Fu, Zhuoyang Zhang, Yecheng Wu, Zhaoshuo Li, Qianli Ma, Song Han, Chelsea Finn, Ankur Handa, Tsung-Yi Lin, Gordon Wetzstein, Ming-Yu Liu, and Donglai Xiang. CoT-VLA: Visual chain-of-thought reasoning for vision-language-action models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2

[55] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Proceedings of Robotics: Science and Systems*, 2023. 5, 6

[56] Hongyan Zhi, Peihao Chen, Siyuan Zhou, Yubo Dong, Quanxi Wu, Lei Han, and Mingkui Tan. 3DFlowAction: Learning cross-embodiment manipulation from 3D flow world model. *arXiv preprint arXiv:2506.06199*, 2025. 2

[57] Chuning Zhu, Raymond Yu, Siyuan Feng, Benjamin Burchfiel, Paarth Shah, and Abhishek Gupta. Unified world models: Coupling video and action diffusion for pretraining on large robotic datasets. In *Proceedings of Robotics: Science and Systems*, 2025. 2

[58] Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Kevin Lin, Abhiram Maddukuri, Soroush Nasiriany, and Yifeng Zhu. robosuite: A modular simulation framework and benchmark for robot learning. *arXiv preprint arXiv:2009.12293*, 2020. 5